

Chenxi Lin

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EDUCATION

Cornell University, College of Engineering Ithaca, NY
Bachelor of Science in Biomedical Engineering May 2028

PROFESSIONAL EXPERIENCE

Joey ELM, Founder, Ithaca, NY June 2026 – Present

- Built joey ELM, an AI research assistant for wet labs, now used by 10+ student researchers at Cornell.
- Gave the agent 16 tools on a Claude tool-use loop with write access to notebook entries, reagent counts, equipment bookings, and to-dos, replacing five separate systems with plain-language requests.
- Wired agent answers to live, lab-scoped searches of OpenAlex, Semantic Scholar, and the researcher's own notebook.

Lee Lab, Cornell University, Research Assistant, Ithaca, NY May 2025 – Present

- Prepare 300+ NGM plates weekly, including custom auxin and RNAi plates, for high-throughput *C. elegans* lifespan and stress-resistance studies.
- Train and manage two undergraduates on plate production and maintain general lab stocks including LB media, antibiotics, and molecular biology buffers.

Roeder Lab, Cornell University, Undergraduate Researcher, Ithaca, NY December 2024 – Present

- Built a MATLAB biomechanical model of plant-to-animal cell transformation with tunable cell wall thickness, shape, and osmolyte concentration, then used it to show that geometry drives growth behavior while thickness and osmolyte concentration only scale it.
- Validated the model against multiple large experimental datasets with curve fitting and stochastic modeling, which improved its predictions of stress-strain and osmoregulatory behavior.
- Optimized protoplast isolation and growth protocols and tracked morphology in real time by live-cell confocal microscopy to validate the model, in collaboration with the Odde Lab (Minnesota) and Boudaoud Lab (École Polytechnique) under the NSF URoL:EN grant.
- Running laser ablation experiments on Arabidopsis sepals to test causes of epidermis to buckling, imaging the ablated region by confocal microscopy and quantifying cell shape change through segmentation.

Blenis Lab, Weill Cornell Medicine, Undergraduate Researcher, New York, NY July 2022 – August 2024

- Led an independent project showing elevated BCAAs levels increased prostate cancer metastasis, and demonstrated experimentally that low-dose TGF- β induces EMT in lung cancer.
- Utilized a wide range of laboratory techniques, including proteomics, metabolomics, gel electrophoresis, 3D and 2D tissue culture, and CRISPR.

PROJECTS

LIVECell U-Net Segmentation, PyTorch, Python, Streamlit, scikit-image, NumPy August 2026

- Built and scaled a U-Net segmentation pipeline on the LIVECell benchmark (5,387 images, 8 cancer cell lines), converting COCO annotations into cached instance masks across official train/validation/test splits.
- Replaced binary segmentation with a three-class interior/boundary model using distance-weighted border loss and seeded watershed, improving CTC SEG from ~ 0.55 to ~ 0.72 while achieving 0.92 Dice and 0.85 IoU.
- Developed instance-level evaluation using CTC SEG, detection F1, merge/split errors, COCO AP/AFNR, and size-stratified recall to capture failures hidden by aggregate Dice.

EXTRACURRICULAR EXPERIENCE

Cornell Maker Club, Member, Ithaca NY January 2026 – Present

- Use CAD, laser cutting, and FDM printing for personal projects and help other members troubleshoot equipment.

Cornell Undergraduate Research Board, Student Mentor, Ithaca NY March 2024 – Present

- Mentored cohorts of 6–10 undergraduate students, providing guidance on research involvement within the field.

HONORS AND AWARDS

Cornell Summer Experience Grant June 2025

- Awarded a \$5,000 competitive Cornell Summer Experience Grant to support summer research.

APIA/United Health Foundation Scholarship May 2024

- Awarded a \$10,000 scholarship in recognition as part of the Diverse Scholars Initiative

SKILLS

- **Skills:** MATLAB, Python, R, SQL, Ansys, MorphoGraphX, Fiji/ImageJ, Mandarin (Native)